

MACHINE LEARNING CNN CLASSIFIER

Project Overview

Machine-learning computer vision pipeline trained on a custom dataset to perform binary and multi-class image classification (Cat vs Non-Cat detection).

Problem Statement: Demonstrating end-to-end ML model training, evaluation, and web deployment with custom dataset pipelines.

Solution Implemented: Trained a deep Convolutional Neural Network (CNN) in TensorFlow/Keras and served inference via a Flask REST API.

My Technical Role: Machine Learning Specialist

Main Features & Technical Stack

Core Features:

- Custom dataset ingestion and data augmentation pipeline
- Deep CNN architecture trained with Keras/TensorFlow
- Flask REST API serving real-time image evaluation predictions

Technologies Used:

Python, TensorFlow, Keras, NumPy, OpenCV, Flask, HTML, CSS, JavaScript

Architecture Flow:

Web UI → Flask Gateway → NumPy Preprocessor → TensorFlow CNN → Render Host

Technical Challenge & Results

Key Engineering Challenge: Applied Dropout layers and L2 regularization to prevent overfitting on limited training data.

Validated Results: Achieved over 90% classification accuracy on evaluation benchmarks.

Live Demo: <https://github.com/prashant-singh-78/image.ml.git> | **GitHub:** <https://github.com/prashant-singh-78/image.ml.git> | **Portfolio:** prashant08.vercel.app